

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. EE Sem – VII Examination, 2013-14
EE - 402 Electrical power system IV

Date: 14/11/2013, Thursday

Time: 10 am to 1 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary

SECTION – I

- Q-1 A. Fig.1 shows a three bus system. The series impedance and shunt admittance of each line is $0.026+j0.11$ pu and $+j0.04$ pu respectively. The bus specification and power input etc. at the buses are as under. Determine the values of load angles of bus 2 and 3 and voltage of bus 3 at the end of first iteration by using either NR method OR FDLF method.

Bus	P_i	Q_i	P_D (pu)	Q_D (pu)	Bus voltage
1 (Slack)	----	----	1	0.5	$V=1.03$ pu $\delta=0$ degree
2 (PV bus)	1.5	----	----	----	$V=1.03$
3 (PQ bus)	---	----	1.2	0.5	----

[14]

- Q-2 A. Write the disadvantages of GS method for load flow analysis.

[6]

A power system has two generators and their incremental fuel cost functions are given as $(6pg_1+1000)$ Rs./MWhr and $(5pg_2+1500)$ Rs. /MWhr. Incremental

- B. transmission losses $(\frac{\partial P_L}{\partial P_g})$ for both generators are 0.05 and 0.1 respectively. Find

[8]

out the optimal values of real power output of both generators using economic load dispatch. Total load demand is 600MW. Loss(PL)= $0.05pg_1+0.1pg_2$.

OR

- Q-2 A. Explain the algorithmic steps of Lamda Iterative method.

[6]

- B. Define economic load dispatch and derive exact coordination equation.

[8]

- Q-3 Explain the classification of buses.

[7]

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SECTION - II

- Q-4 A. For the network shown in figure 2, find out Z_{bus} using various types of modifications. [8]
 B. Explain type 1 and 2 modifications for fault analysis. [6]
- Q-5 A. Explain turbine speed governing system to control frequency of the system. [8]
 B. Explain Series and shunt compensation in transmission system with necessary phasor diagram. [6]
- OR
- Q-5 A. Derive Wave equation of Traveling wave on transmission line. [8]
 B. Explain mechanism of lightning stroke and its characteristics. [6]
- Q-6 Discuss different types of protection against surges. [7]

